

Oksendal 3.2

Prove directly from the definition of Itô integrals that:

$$\int_0^t B_s^2 dB_s = \frac{1}{3} B_t^3 - \int_0^t B_s ds$$

Solution

To prove this directly from the definition, we use the property of Riemann-Stieltjes sums (specifically, the Itô isometry approach) by partitioning the time interval.

1. Set the Partition

Let $0 = t_0 < t_1 < \dots < t_n = t$ be a partition of $[0, t]$, where $\Delta t_i = t_{i+1} - t_i$. By the definition of the Itô integral:

$$\int_0^t B_s^2 dB_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} B_{t_i}^2 (B_{t_{i+1}} - B_{t_i})$$

2. Use the Algebraic Identity

To relate the sum to B_t^3 , we use the identity for $a^3 - b^3$:

$$(B_{t_{i+1}} - B_{t_i})^3 = B_{t_{i+1}}^3 - B_{t_i}^3 - 3B_{t_i}^2(B_{t_{i+1}} - B_{t_i}) - 3B_{t_i}(B_{t_{i+1}} - B_{t_i})^2$$

Rearranging this to isolate the term $B_{t_i}^2(B_{t_{i+1}} - B_{t_i})$ (which matches our integral sum):

$$B_{t_i}^2(B_{t_{i+1}} - B_{t_i}) = \frac{1}{3} \left[(B_{t_{i+1}}^3 - B_{t_i}^3) - 3B_{t_i}(B_{t_{i+1}} - B_{t_i})^2 - (B_{t_{i+1}} - B_{t_i})^3 \right]$$

3. Summing Over the Partition

Summing both sides from $i = 0$ to $n - 1$:

$$\sum B_{t_i}^2 \Delta B_i = \frac{1}{3} \sum (B_{t_{i+1}}^3 - B_{t_i}^3) - \sum B_{t_i} (\Delta B_i)^2 - \frac{1}{3} \sum (\Delta B_i)^3$$

4. Evaluate the Limits

Now, we take the limit as the mesh size goes to zero ($n \rightarrow \infty$):

- Telescoping Sum: The first term is a telescoping sum that simplifies to the endpoints:

$$\frac{1}{3} \sum_{i=0}^{n-1} (B_{t_{i+1}}^3 - B_{t_i}^3) = \frac{1}{3} (B_t^3 - B_0^3) = \frac{1}{3} B_t^3$$

- Quadratic Variation Term: From the properties of Brownian motion, $(\Delta B_i)^2 \approx \Delta t_i$. Specifically, $\sum B_{t_i}(\Delta B_i)^2$ converges in L^2 to the Riemann integral:

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} B_{t_i} (B_{t_{i+1}} - B_{t_i})^2 = \int_0^t B_s ds$$

- Higher Order Term: The term $\sum (\Delta B_i)^3$ vanishes in the limit because $(\Delta B_i)^3$ is of order $(\Delta t)^{3/2}$:

$$\lim_{n \rightarrow \infty} \frac{1}{3} \sum_{i=0}^{n-1} (B_{t_{i+1}} - B_{t_i})^3 = 0$$

5. Final Result

Substituting these limits back into our summed equation:

$$\int_0^t B_s^2 dB_s = \frac{1}{3} B_t^3 - \int_0^t B_s ds$$

Klebner 4.3

Let X_n be a Gaussian sequence convergent in distribution to X . Show that the distribution of X is either Normal or degenerate. Deduce that if $E[X_n] \rightarrow \mu$ and $Var(X_n) \rightarrow \sigma^2 > 0$ then the limit is $N(\mu, \sigma^2)$. Since convergence in probability implies convergence in distribution, deduce convergence of Itô integrals of simple non-random processes to a Gaussian limit.

Solution

We can use the characteristic function to show that X is Normal or degenerate. Let $X \sim N(\mu_n, \sigma_n^2)$, then the characteristic function of X_n is the following:

$$\phi_{X_n}(t) = E[e^{itX_n}] = \exp\left(it\mu_n - \frac{1}{2}\sigma_n^2 t^2\right)$$

If $X_n \xrightarrow{d} X$, then by Lévy's Continuity Theorem, $\phi_{X_n}(t) \rightarrow \phi_X(t)$ for all t , where $\phi_X(t)$ is continuous at $t = 0$. For this limit to exist:

1. The sequence of means μ_n must converge to some μ .
2. The sequence of variances σ_n^2 must converge to some $\sigma^2 \geq 0$.

If we take the limit we get the following:

$$\phi_X(t) = \lim_{n \rightarrow \infty} \exp\left(it\mu_n - \frac{1}{2}\sigma_n^2 t^2\right) = \exp\left(it\mu - \frac{1}{2}\sigma^2 t^2\right)$$

- Case 1: If $\sigma^2 > 0$, this is the characteristic function of $N(\mu, \sigma^2)$.
- Case 2: If $\sigma^2 = 0$, then $\phi_X(t) = e^{it\mu}$, which is the characteristic function of a degenerate distribution (a constant $X = \mu$ with probability 1).

Since we are given specifically that $E[X_n] \rightarrow \mu$ and $Var(X_n) \rightarrow \sigma^2 > 0$, we can substitute these directly into the characteristic function from above. Since σ^2 is only positive, the limit cannot be degenerate, confirming $X \sim N(\mu, \sigma^2)$.

Application to Itô Integrals

For a deterministic (non-random) step function $f(s) = \sum_{i=0}^{n-1} a_i \mathbf{1}_{[t_i, t_{i+1})}$, the Itô integral is:

$$I(f) = \sum_{i=0}^{n-1} a_i (B_{t_{i+1}} - B_{t_i})$$

Each increment $(B_{t_{i+1}} - B_{t_i})$ is a Gaussian random variable. Since the increments are independent and the coefficients a_i are constants, $I(f)$ is a linear combination of independent Gaussians, which is itself Gaussian.

For a general non-random process $g \in L^2([0, t])$, we define the integral as the limit in probability of a sequence of simple processes f_n :

$$\int_0^t g(s)dB_s = \lim_{n \rightarrow \infty} I(f_n)$$

The Guassian Limit:

1. Each $I(f_n)$ is a Gaussian sequence X_n .
2. Convergence in probability implies convergence in distribution.
3. From our proof in the first part, the distribution of the limit must be Gaussian.
4. Using Itô Isometry:
 - $E[I(f_n)] = 0$
 - $Var(I(f_n)) = \int_0^t f_n(s)^2 ds \rightarrow \int_0^t g(s)^2 ds$

The Itô integral of a non-random process results in a Gaussian random variable $N(0, \int_0^t g(s)^2 ds)$.